

مسألة 1.6

(1.6)

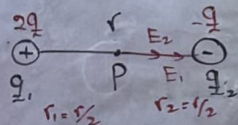
A proton moves in $E = 500 \text{ N}$ at $(+x)$. The acceleration

$$a = \frac{F}{m} = \frac{qE}{m} = \frac{1.6 \times 10^{-19} \times 500}{1.67 \times 10^{-27}}$$

$$a = 4.8 \times 10^{10} \text{ m/s}^2 (+x)$$

(1.7)

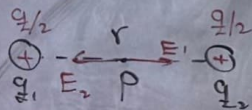
q_1 is two times of q_2 , if the two charges brought into contact and then separated to the same distance. The new electric field at the middle.



$$E = E_1 + E_2 = k_e \frac{q_1}{r_1^2} + k_e \frac{q_2}{r_2^2} = k_e \frac{2q}{(r/2)^2} + k_e \frac{q}{(r/2)^2}$$

$$E = 8k_e \frac{q}{r^2} + 4k_e \frac{q}{r^2} = 12k_e \frac{q}{r^2}$$

$$24 = 12k_e \frac{q}{r^2} \rightarrow \boxed{2 = k_e \frac{q}{r^2}}$$



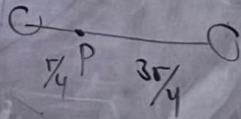
Contact $\rightarrow 2q - q = q$

Separate $q_1 = q_2 = q/2$

$$E_{\text{new}} = E_1 - E_2 = k_e \frac{q/2}{(r/2)^2} - k_e \frac{q/2}{(r/2)^2} = 0$$

$$= k_e \frac{3/2 q}{r^2} - k_e \frac{2/6 q}{r^2}$$

$$\boxed{k_e \frac{q}{r^2} \left(\frac{3}{2} - \frac{2}{6} \right)}$$



$$\frac{(3/4 r)^2}{k_e \frac{q/2}{r^2}} = \frac{8/9 k_e q}{r^2}$$

المسألة رقم 1.8

$q_1 = 6 \text{ nC}$ Placed $(30 \text{ cm}, 0)$

$q_2 = -3 \text{ nC}$ Placed $(0, -10 \text{ cm})$

The electric field at Point P placed at origin.

$$E_1 = k \frac{q_1}{r_1^2} = 9 \times 10^9 \times \frac{6 \times 10^{-9}}{0.3^2} = 600 \text{ N/C}$$

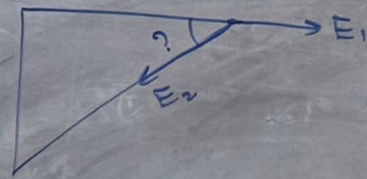
$$E_2 = k \frac{q_2}{r_2^2} = 9 \times 10^9 \times \frac{3 \times 10^{-9}}{0.1^2} = 2700 \text{ N/C}$$

$$E_x = -600$$

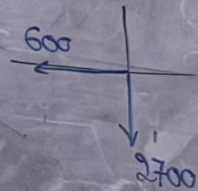
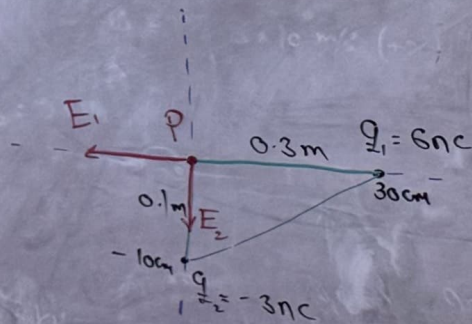
$$E_y = -2700$$

$$E = \sqrt{600^2 + 2700^2}$$

$$\theta = \tan^{-1}\left(\frac{2700}{600}\right) = \theta \rightarrow \theta_{\text{rx}} = 180^\circ + \theta$$



$$\theta = \tan^{-1}\left(\frac{0.1}{0.3}\right) = \theta$$



الزوايا

مسألة 1.8

$q_1 = 6 \text{ nC}$ Placed origin

$q_2 = -3 \text{ nC}$ Placed $(0, 40 \text{ cm})$

The electric field at Point P placed at $(30 \text{ cm}, 0)$

$$E_1 = k_e \frac{q_1}{r_1^2} = 9 \times 10^9 \times \frac{6 \times 10^{-9}}{0.3^2} = 600 \text{ N/C}$$

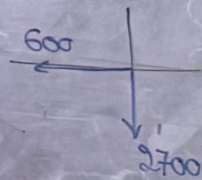
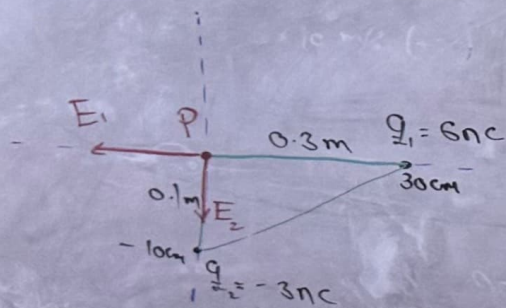
$$E_2 = k_e \frac{q_2}{r_2^2} = 9 \times 10^9 \times \frac{3 \times 10^{-9}}{0.1^2} = 2700 \text{ N/C}$$

$$E_x = -600$$

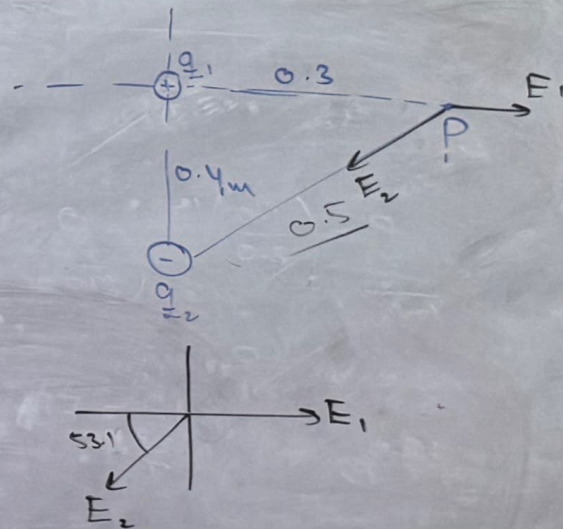
$$E_y = -2700$$

$$E = \sqrt{600^2 + 2700^2}$$

$$\theta = \tan^{-1}\left(\frac{2700}{-600}\right) \rightarrow \theta_{tx} = 180^\circ + 0^\circ$$



36.87
63.13



بسم الله الرحمن الرحيم

(1.8)

$q_1 = 6 \text{ nC}$ Placed origin

$q_2 = -3 \text{ nC}$ Placed $(0, 40 \text{ cm})$

The electric field at Point P placed at $(30 \text{ cm}, 0)$

بسم الله الرحمن الرحيم

(1.8)

$q_1 = 6 \text{ nC}$ Placed origin

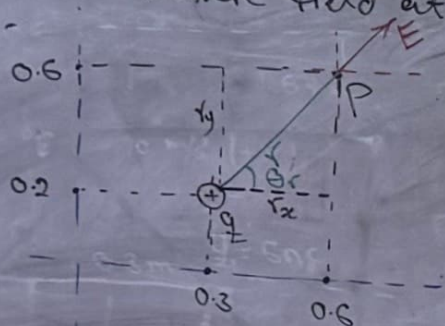
$q_2 = -3 \text{ nC}$ Placed $(0, 40 \text{ cm})$

The electric field at Point P placed at $(-30 \text{ cm}, 0)$

1.9

$q_1 = 5 \mu\text{C}$ placed at $(0.3\text{m}, 0.2\text{m})$

The electric field at Point P placed at $(0.6\text{m}, 0.6\text{m})$



a) $E = k \frac{q}{r^2} = 9 \times 10^9 \times \frac{5 \times 10^{-6}}{0.5^2} = 180000 \text{ N/C}$ at $\theta = 53.1^\circ$

b) The electric force on $q_2 = 2 \mu\text{C}$ is:

$F = q_2 E = 2 \times 10^{-6} \times 180000 = 0.36 \text{ N}$ at $\theta = 180 + 53.1 = 233.1^\circ$

$r_x = 0.6 - 0.3 = 0.3$

$r_y = 0.6 - 0.2 = 0.4$

$r = \sqrt{0.3^2 + 0.4^2} = 0.5 \text{ m}$

$\theta = \tan^{-1}\left(\frac{r_y}{r_x}\right) = \tan^{-1}\left(\frac{0.4}{0.3}\right) = 53.1^\circ$

السبب لا السبب ثابت
فإن القوة معاكسة للحال

$$(0.6, 0.3)$$

$$(-0.3, -0.1)$$

$$x = 0.9$$

$$y = 0.4$$

m)

at $\theta = 53.1$

o:

at $\theta = 180 + 53.1 = 233.1$

بسم الله الرحمن الرحيم

1.9

$q_1 = 5 \mu\text{C}$ placed at $(0.6\text{m}, 0.6\text{m})$

The electric field at Point P placed at $(0.3\text{m}, 0.2\text{m})$

$(0.6, 0.3)$

$(-0.3, -0.1)$

$x = 0.9$

$y = 0.4$

(2)

$q = -5 \mu\text{C}$ placed at $(0.1 \text{ m}, 0.6 \text{ m})$

The electric field at point P placed at $(-0.2 \text{ m}, 0.2 \text{ m})$

$$(0.6, 0.3)$$

$$(-0.3, -0.1)$$

$$x = 0.9$$

$$y = 0.4$$